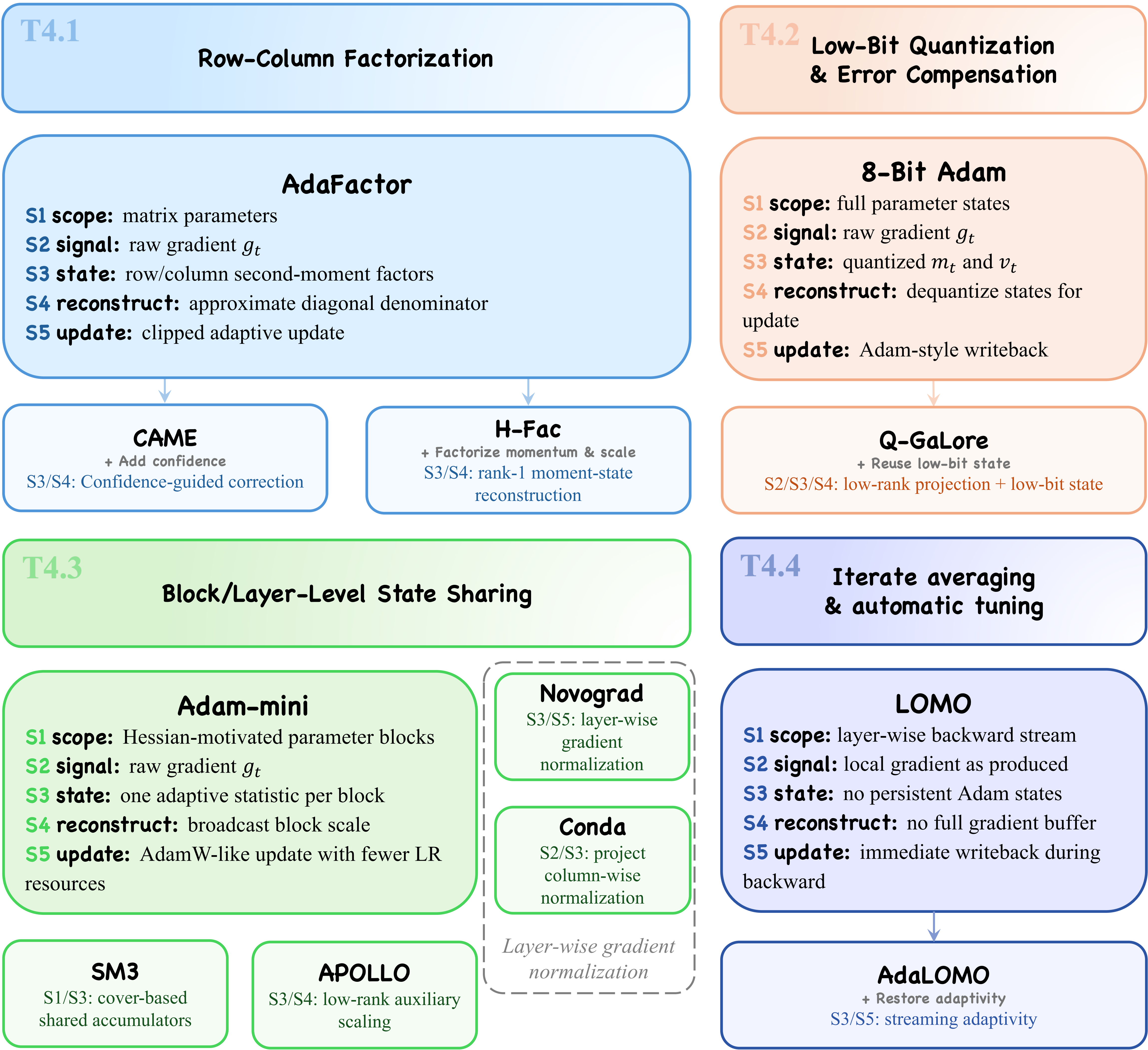
T4.1
Row-Column Factorization
AdaFactor
S1 scope: matrix parameters
S2 signal: raw gradient g_t
S3 state: row/column second-moment factors
S4 reconstruct: approximate diagonal denominator
S5 update: clipped adaptive update
CAME
+ Add confidence
S3/S4: Confidence-guided correction
H-Fac
+ Factorize momentum & scale
S3/S4: rank-1 moment-state reconstruction
T4.2
Low-Bit Quantization & Error Compensation
8-Bit Adam
S1 scope: full parameter states
S2 signal: raw gradient g_t
S3 state: quantized m_t and v_t
S4 reconstruct: dequantize states for update
S5 update: Adam-style writeback
Q-GaLore
+ Reuse low-bit state
S2/S3/S4: low-rank projection + low-bit state
T4.3
Block/Layer-Level State Sharing
Adam-mini
S1 scope: Hessian-motivated parameter blocks
S2 signal: raw gradient g_t
S3 state: one adaptive statistic per block
S4 reconstruct: broadcast block scale
S5 update: AdamW-like update with fewer LR resources
Novograd
S3/S5: layer-wise gradient normalization
Conda
S2/S3: project column-wise normalization
Layer-wise gradient normalization
SM3
S1/S3: cover-based shared accumulators
APOLLO
S3/S4: low-rank auxiliary scaling
T4.4
Iterate averaging & automatic tuning
LOMO
S1 scope: layer-wise backward stream
S2 signal: local gradient as produced
S3 state: no persistent Adam states
S4 reconstruct: no full gradient buffer
S5 update: immediate writeback during backward
AdaLOMO
+ Restore adaptivity
S3/S5: streaming adaptivity